Ira A. Fulton Schools of Engineering
Arizona State University

CSE 579: Knowledge Representation and Reasoning

More Challenging Answer Set Programming

Purpose

In this second programming assignment, you will again use clingo, this time to solve more complex problems, specifically four problems involving actions and effects. You will use ASP to formulate transition systems involving actions and effects, decisions, etc. Full instructions and assignment details follow

Objectives

Learners will be able to:

- Become more familiar with the Blocks World problem by applying different constraints.
- Gain experience in implementing various constraints on both serializable and optimal plans in problems 3 & 4.

Technology Requirements

Clingo

Project Description

This assignment requires you to complete a set of four more challenging clingo programming problems. Please follow the directions below to complete the assignment

Directions

1. Download the assignment template and the problems from your course “Assignment Overviews and Resources” page. Read all of the problems and review the evaluation criteria section that follows. NOTE: All of the problems are provided in the attached PDF for initial review and reference as you work in clingo.

    a. CSE 579_GradedAssignment2_Template

b. CSE 579_GradedAssignment2_Problems

2. Access clingo on your local machine.

    a. You must use clingo to complete this assignment. You also must complete this assignment using clingo on your **local machine**, because the online version of the tool will not include the command line, which you are required to submit to complete the assignment. The installation guide is available at https://github.com/zhunyoung/clingoTutorial Opens in a new tab .

3. Complete the problems.

    a. Complete problems 1 through 4 using the attached template (provided above) to collect the information required for each problem. Note that some hints are also given in the template to help you to complete these 4 problems. Once you have filled in all the tables in the template, save it as a pdf file named "Your name_pa2.pdf".

Submission Directions for Project Deliverables

You are given an unlimited number of attempts to submit your best work. The number of attempts is given to anticipate any submission errors you may have in regards to properly submitting your best work within the deadline (e.g., accidentally submitting the wrong paper). It is not meant for you to receive multiple rounds of feedback and then one (1) final submission. Only your most recent submission will be assessed.

You must submit your More Challenging Answer Set Programming Assignment deliverable in this submission space. Learners may not email or use other means to submit any assignment or project for review, including feedback, and grading.

The More Challenging Answer Set Programming Assignment includes one (1) deliverable:

- **Solution PDF:** Submit your completed assignment using the assignment template. This document is available on the Project Overview page in your course. Your file upload must follow the correct naming convention: Your Name_CSE 579_More Challenging Answer Set Programming Assignment.

To submit your work follow the directions below:

1. Navigate to your course and click on the "**Submission: More Challenging Answer Set Programming**" submission space
2. Click "**Start Attempt**" in the upper right corner
3. Click "**Upload File**" and add your template pdf document

a. When finished, click “**Submit Assignment**”
b. If needed: to resubmit the assignment
    i. Click “**New Attempt**” and add your file submission again
    ii. When finished, click “**Submit Assignment**”

Evaluation

This assignment is staff-graded. There are a total of 4 problems for a total of 100 points.

Rubric

Please review the rubric found on the submission page for this assignment.